

## Italian\_exam\_A002\_ORD24-English

### Italian Maturità — Esame di Stato, Mathematics (A002)

**Ordinary session 2024 — Second written test** Source: *Ministero dell'istruzione e del merito*

Valid for the Scientific Lyceum (indirizzi LI02, LI03, LI15, LI1S, LI22, LI23, LI31, LI32, LIA2, LIAO, LIB2, LIC2, LID2, LII2, LII3, LII4, LIIS, LIS2, EA02, EA10).

**Instructions:** The candidate must solve **one** of the two problems and answer **4** of the questions in the questionnaire.

#### Problem 1

Consider  $f_{a,b}(x) = a + b \cdot x$ .

**(a)** Find the values of the parameters such that the line  $(t): (7x + y - 12 = 0)$  is tangent to the graph of  $(f_{a,b}(x))$  at its point  $(P)$  of abscissa  $(x=1)$ .

From now on, set  $(a=1)$  and  $(b=4)$ .

**(b)** Study the function  $[f(x) = ]$  and sketch its graph  $()$ . Write the equation of the other tangent line to the curve  $()$  that passes through  $(P)$ .

**(c)** As the real parameter  $(m)$  varies, determine the number of intersections between the line  $(y - 5 = m(x - 1))$  and the curve  $()$ .

**(d)** Let  $(S(k))$ , with  $(k > 0)$ , denote the area of the bounded planar region delimited by the curve  $()$ , its oblique asymptote, the line  $(t)$ , and the line  $(x = k)$ . Compute  $[ \lim_{k \rightarrow +\infty} S(k), ]$  and give a geometric interpretation of the result.

#### Problem 2

*“At the beginning and at the end, we have mystery. [...] Mathematics brings us close to this mystery, without penetrating it.” — E. De Giorgi*

Consider the family of functions  $[f_n(x) = -x^n, n > 1, a, b, a < 0. ]$

**(a)** Verify that, for every  $(n)$ , the function  $(f_n)$  is not differentiable at  $(x=0)$ . Determine the value of  $(n)$  for which the graph of  $(f_n)$  has a corner point. For suitable values of  $(a, b)$ , the graph  $()$  shown in the figure represents the function  $[f_2(x) = |x| - 1. ]$  Determine  $(a)$  and  $(b)$ , given that  $(f_2)$  is defined on  $([-1; 1])$  and that its graph is symmetric about the  $(y)$ -axis.

From now on, set  $(a=-1)$  and  $(b=0)$ .

**(b)** Study the function  $[g(x) = |x| + 1, ]$  verifying that it is not differentiable at the endpoints of the domain and at  $(x=0)$ . Denote its graph by  $()$  and sketch the curve  $(=)$ .

**(c)** The line  $(r): (x=k)$ , with  $(-1 < k < 1)$ , intersects  $()$  at the points  $(P)$  and  $(Q)$ . Show that the length of the segment  $(PQ)$  is maximum when  $(r)$  is an axis of symmetry of  $()$ .

(d) Verify that  $[H(x) = ((x)+x)]$  is an antiderivative of  $(h(x) = )$ . By any method, compute the area of the bounded region enclosed by  $()$ .

*“A mathematician’s patterns, like the painter’s or the poet’s, must be beautiful: the ideas, like the colours or the words, must fit together in a harmonious way. Beauty is the first test: there is no permanent place in the world for ugly mathematics.” — G. H. Hardy*

## Questionnaire (Quesiti)

1. A triangle (ABC) has a right angle at (B). Prove that the triangle is isosceles if and only if the altitude (BH) to the hypotenuse is congruent to half the hypotenuse.
2. A loaded coin showing heads with probability (p) is tossed 5 times.
  - What is the probability of obtaining exactly 2 heads?
  - For which value of (p) is this probability maximum?

3. In space with an orthogonal Cartesian frame (Oxyz), consider the plane  $(3x-2y+5=0)$ .

- Find the coordinates of the point (H), the orthogonal projection of  $(P(4,2,1))$  onto the plane  $()$ .
- Find the intersection of the line  $(s)$

$$\begin{cases} x - y + 1 = 0 \\ z - 2 = 0 \end{cases}$$

$()$  with the plane  $()$ .

4. Prove that the equation  $(x^3+x-x=0)$  has a unique positive solution.
5. Determine the fourth-degree polynomial  $(y=p(x))$  whose graph satisfies the following:
  - it is tangent to the (x)-axis at the origin;
  - it passes through the point  $((1,0))$ ;
  - it has a stationary point at  $((2,-2))$ .
6. Consider the integral function  $[F(x) = \int_a^x f(t)dt, x \in ]a, b[ ]$  where (a) is a positive real parameter. Determine the greatest value of (a) such that  $(F(2) = -)$ .
7. On 5 July, Earth will reach the aphelion, the point of its orbit at the maximum distance from the Sun, equal to about  $(1.52 \times 10^{11})$  m. The perihelion is the point at the minimum distance from the Sun, about  $(1.47 \times 10^{11})$  m. In a suitable reference frame, determine the equation that represents Earth’s orbit around the Sun.
8. Carlo Emilio Gadda writes in one of the stories of *L’Adalgisa – Disegni milanesi*: > “The service rooms, the bath, the corridors, the antechamber and one of the two lavatories were paved with small red tiles: hexagonal [...]. The apothem of those tiles measured 5.196 centimetres: whereas the radius of the circumscribed circle reached 60 millimetres.”

State the exact relation between the radius of the circumscribed circle and the apothem (the radius of the inscribed circle) for a regular hexagon. Verify the result against the measurements indicated by the writer.

Explain why a plane can be tiled using regular hexagonal tiles that are all congruent. Which other regular polygons, all congruent to one another, can tile a plane? Justify the answer.

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**Maximum duration:** 6 hours. Scientific or graphing calculators are allowed, provided they have no symbolic algebraic processing capability and no internet connection.